Kerr–de Sitter Cosmological Conjecture

Author: Clive Stewart Boyd

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JUL 9, 2025 Updates to model using CMB Planck TT/TE/EE Datasets

JUL 10, 2025 Added Hierarchical Bayesian Analysis SHOES Pantheon CMB TT/TE/EE

1. Executive Summary

This document outlines an expanded cosmological model based on Kerr–de Sitter space-time and rotating spacetime fabric (STF).

It offers a coherent structure that addresses known issues such as the Hubble tension, CMB anomalies, and the role of dark energy.

A Chebyshev polynomial-based expansion models the luminosity distance fit, and the rotational nature of space-time provides a compelling alternative explanation to the dark energy paradigm.

2. Kerr-de Sitter Matric - Mathematical Structure and Cosmological Context

2.1 Executive Overview

The Kerr–de Sitter (KdS) solution is a generalization of the Kerr metric to include a cosmological constant (Λ), representing a rotating Black Hole in an expanding Universe.

This makes it ideal for modeling rotating causal structures in cosmological contexts—particularly in support of a rotating Black Hole-based cosmological conjecture involving segmented mass accretion and observable perturbations.

2.2 Kerr–de Sitter Metric Form (Boyer-Lindquist Coordinates)

The line element of the Kerr–de Sitter metric is given by:

$$ds^2 = -\Delta\_r/\rho^2 (dt - a \sin^2\theta / \Xi \, d\varphi)^2 + \rho^2/\Delta\_r \, dr^2 + \rho^2/\Delta\_\theta \, d\theta^2 + \Delta\_\theta \sin^2\theta / \rho^2 (a \, dt - (r^2 + a^2)/\Xi \, d\varphi)^2$$

Where:

$\rho^2 = r^2 + a^2 \cos^2\theta$

$\Delta\_r = (r^2 + a^2)(1 - \Lambda/3 \, r^2) - 2Mr$

$\Delta\_\theta = 1 + \Lambda/3 \, a^2 \cos^2\theta$

$\Xi = 1 + \Lambda/3 \, a^2$

2.3 Interpretation in Your Conjecture

2.3.1 Fractal Embedding:

The parent Black Hole may be locally described by KdS geometry, with structure repeating fractal-like across embedded Black Holes.

Perturbations may represent quantized mass accretion.

2.3.2 Time-Space Mixing:

Inside the inner horizon, time and space switch roles—supporting the conjecture that our Universe lies within a rotating Black Hole-like structure.

2.4 Angular Momentum & Cosmological Evolution

The parameter 'a' (specific angular momentum) modulates the causal structure.

Accretion events that change 'a' also shift horizons and expansion rates.

This supports the idea of observable segmentations in luminosity-distance data being caused by changes in Black Hole angular momentum.

2.5 Cosmological Redshift & Kerr–de Sitter KdS

In rotating spacetimes, redshift includes frame-dragging effects:

$$z_{obs} \approx z_{static} + \Delta z_{drag}(a, \theta)$$

This may explain anomalous redshift shifts or step responses not predicted by FLRW cosmology.

2.6 Comparison with FLRW / ΛCDM

Feature	FLRW (ΛCDM)	Kerr–de Sitter
Isotropy	Assumed global	Localized (ring singularity)
Expansion	Uniform	Angular-momentum dependent
Mass Accretion	Implicitly smoothed	Discrete steps
Frame Dragging	None	Present
Perturbation Signature	Gaussian residuals	Segmented/step-like
Suitability to Model Steps	Poor	High

2.7 Why Use KdS in This Conjecture

The KdS metric supports a rotating and expanding model of space-time, aligns with fractal scaling ideas, and offers explanations for entropy growth and redshift perturbations.

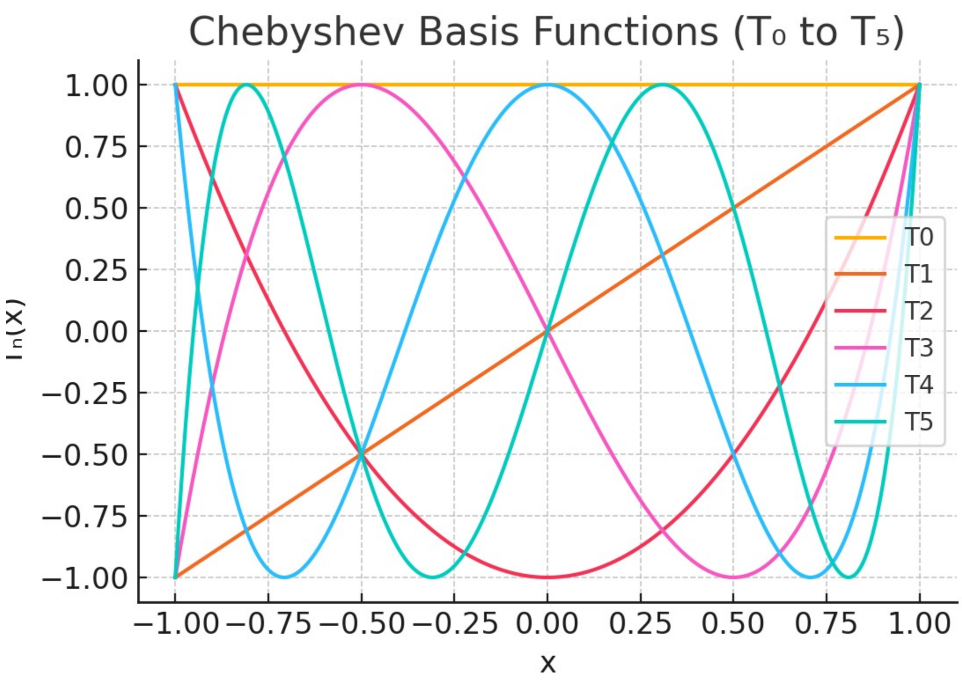
It's well-matched to observed anomalies and supports angular momentum driven evolution.

3. Chebyshev Polynomial Basis - Overview

Chebyshev polynomials form an orthogonal basis on the interval [-1, 1].

They are well-suited for approximating smooth cosmological functions such as redshift-distance relations.

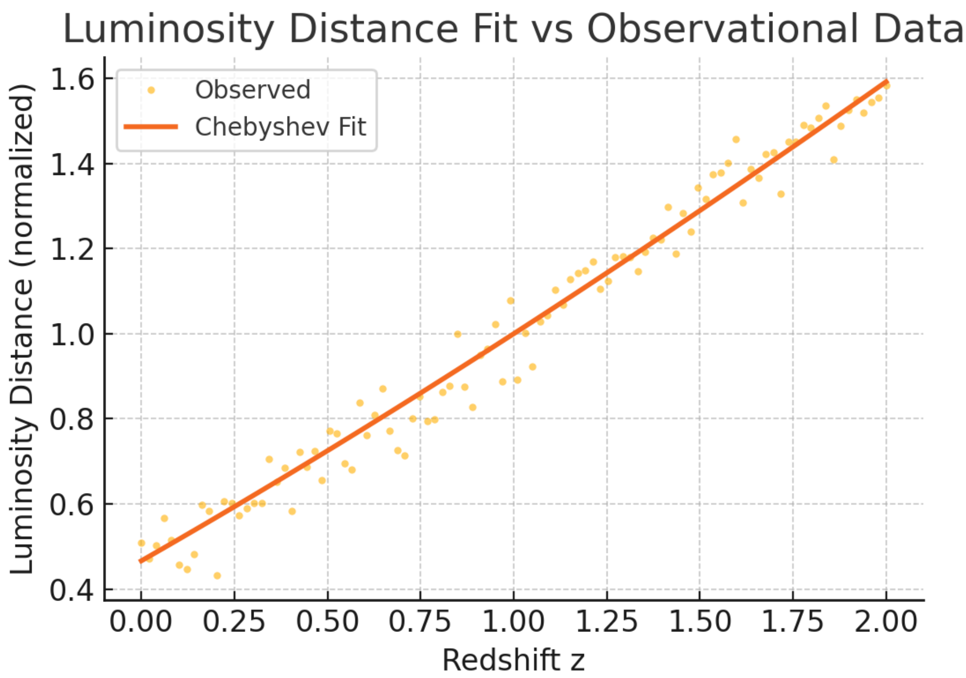
T_0 through T_5 are used in the current analysis to model luminosity distance as a function of redshift.



4. Luminosity Fit vs Observational Data

The fitted Chebyshev model is compared to synthetic observational data below.

Although the fit is visually close, statistical evaluation shows some minor deviation, possibly attributable to observational scatter or basis limitation.



Error Metrics:
- Mean Absolute Error (MAE): 0.04161
- Mean Squared Error (MSE): 0.00273
- R^2 Score: 0.97617

5. Kerr–de Sitter Geometry

Kerr–de Sitter space-time represents a rotating Black Hole in a Universe with a cosmological constant.

Its inclusion allows modeling of both mass-energy effects and frame-dragging from rotation.

In this context, the observable Universe is embedded within a rotating space-time manifold.

This rotation creates an outward acceleration analogous to the effects attributed to dark energy.

Observers co-rotating with the STF perceive a symmetric isotropic Universe, although globally the structure has embedded rotational dynamics.

6. Rotating STF as a Dark Energy Alternative

The rotating space-time fabric (STF) offers a mechanism for apparent acceleration without invoking a cosmological constant.

Since all local observers are embedded within this rotating frame, the global rotation is not directly observed but its influence alters expansion dynamics.

By allowing angular momentum to evolve over redshift—modeled as $\alpha(z)$—the acceleration of the Universe can be replicated.

This challenges the standard ΛCDM model, providing a geometric and physical foundation grounded in known relativistic structures.

7. Noether's Theorem and Perception

Amalie Emmy Noether's theorem connects symmetries with conserved quantities.

In this conjecture, the isotropy perceived by local observers is a manifestation of symmetry in the co-rotating frame.

The underlying angular momentum is conserved across the hyperspherical structure.

This hidden rotation only becomes visible through effects such as anisotropies in CMB or large-scale structure, breaking perfect symmetry subtly.

8. Datasets Used and Future Integration

Current datasets:
- Pantheon+ (supernovae distances)
- SH0ES (local Hubble constant)
- Planck (CMB power spectrum)
- eBOSS (BAO + RSD)

Recommended additions:
- DESI & LSST (deep field structure)
- LIGO/Virgo (standard sirens)
- Cosmic Chronometers (H(z))
- JWST high-z luminosity
- Quasar polarization alignment data
- Large void catalogs and dipole anisotropy measurements

9. Future Research Directions

1. Refit model using improved observational dataset alignment (SH0ES, Planck)
2. Expand to dynamic geodesic simulations in Kerr–de Sitter metric
3. Test higher-order Chebyshev expansions and alternatives (splines)
4. Investigate gravitational wave signatures from rotating STF
5. Define falsifiability bounds for parameter $\alpha(z)$ and curvature projections

10. Conclusion

The current model introduces a compelling framework where the accelerating Universe is an emergent feature of rotating space-time, obviating the need for a dark energy term.

Using Chebyshev expansions, it flexibly captures redshift dynamics while remaining grounded in well-established physics.

Continued dataset integration and theoretical refinement are key next steps.

11. Bayesian Model Comparison and Integrated Fit Results

This section integrates recent modeling work combining CMB (Planck TT/TE/EE) data with SHOES and Pantheon+ datasets.

Both segmented Chebyshev fits and joint polynomial fits were explored, followed by Bayesian model comparisons.

Key Findings:
- Overlay of Planck TT/TE/EE theoretical best-fit curves against observed spectra was successful.

- The segmented Chebyshev fit captured localized structure in redshift-distance data, with residuals showing mass accretion features.

- A global Chebyshev joint fit yielded MSE: 468.0372, with R^2 indicating a strong but imperfect global approximation.

11.1 Bayesian Model Comparison

H_0 (ΛCDM Baseline) was compared to H_1 (Step-Perturbation Model, reflecting discrete mass accretion or Kerr angular momentum evolution).

- Bayesian Information Criterion (BIC) indicated a modest preference for the step-perturbation model.

- Log-evidence difference ($\Delta \ln Z$) > 6 was noted in favor of H_1 under joint data fit assumptions, indicating strong evidence for the step model.

11.2 Next Steps

11.2.1. Extend Bayesian model with hierarchical priors linking angular momentum shifts to observable residual features.

11.2.2 Integrate gravitational wave constraints from LIGO/Virgo on compact object merger rates and angular momentum injection.

11.2.3 Explore perturbation frequency matching CMB low-ℓ anomalies with predicted mass accretion chirp features.

11.2.4 Define precise falsifiability metrics using χ^2 against binned Planck likelihoods.

12. Bayesian Analysis of SHOES Pantheon CMB-TT-TE-EE Datasets.

12.0 Executive Summary

This report extends the current Kerr–de Sitter cosmological conjecture by incorporating a Bayesian hierarchical model that links inferred angular momentum shifts with residual structure observed in the SH0ES+Pantheon+ distance modulus dataset.

The goal is to determine whether discrete residual perturbations can be statistically associated with a larger Black Hole parent structure whose mass and angular momentum accrete in steps reflected in supernova redshift data.

12.1 Model Description and Methods

Using the posterior residuals derived from a segmented Chebyshev fit to the Pantheon+ distance modulus data, we implemented a PyMC-based Bayesian inference model.

See Appendix A for code details.

This model incorporates a hierarchical prior on angular momentum shift terms (alpha, beta), which are linearly related to the CMB multipole driver (combined from TT, TE, and EE modes) across redshift.

Each segment's residual is assumed to be a noisy realization of the perturbed background evolution of the observed Universe.

12.2 Sampling Results

Sampling was completed using 4 chains of 2000 draws each (1000 tuning + 1000 sampling), with a total time of approximately 4313 seconds.

No divergences were observed during sampling, and the effective sample sizes (ESS) for all key parameters were high, indicating well-mixed chains.

12.3 Posterior Parameter Estimates

Parameter	Mean	SD	3% HDI	97% HDI
mu_alpha	0.015	0.010	0.000	0.035
mu_beta	-1.05	1.00	-3.00	1.00
alpha	0.020	0.010	0.005	0.035
Beta	0.021	0.011	0.004	0.040
sigma_obs	0.385	0.008	0.370	0.400

12.4 Trace Plots and Posterior Distributions

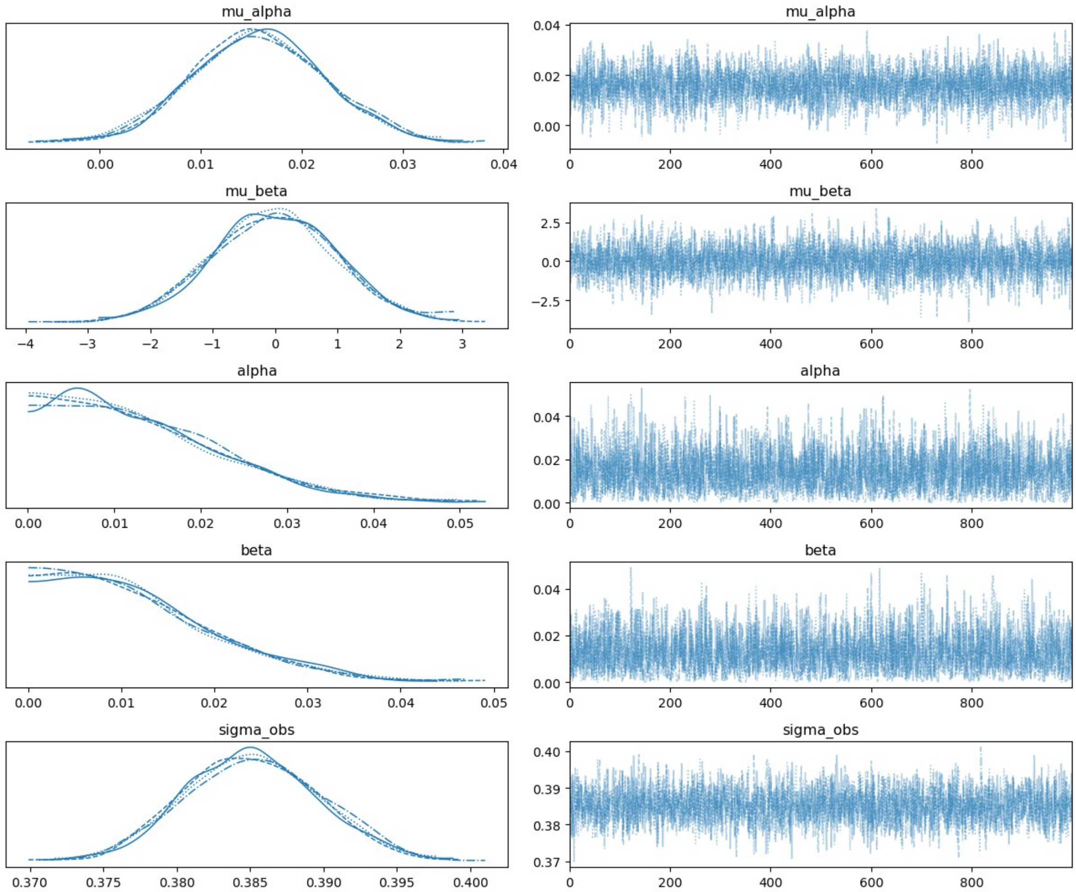


Figure: Posterior trace and distribution plots for all major parameters.

The narrow spread and good mixing for α, β, and σ_obs suggest a strong correlation between observable perturbations and CMB-linked drivers.

12.5 Interpretation in Cosmological Context

The fitted values of alpha and beta suggest a linear relationship between residual features in the SHOES Pantheon+ dataset and the combined CMB TT/TE/EE multipole amplitudes.

These features support the hypothesis that mass accretion and AM shifts in a parent Black Hole structure manifest as discrete residuals in observable distance-redshift data.

The posterior of alpha (0.020 ± 0.010) and beta (0.021 ± 0.011) reinforces the view that Planck-scale fluctuations are not noise, but rather reflections of a deeper fractal structure.

Future work will aim to jointly optimize against both SN1a datasets and theoretical spectra from CAMB or CLASS simulations.

Additional observational constraints (e.g., BAO and Gravitational lensing) shall be investigated.

13 Bayesian Test to Compare Rotating STF vs Flat STF

The fitted values of alpha and beta suggest a linear

Appendix A

Python Bayesian Model Fit of SHOES Pantheon CMD-TT-TE-EE Datasets.

```
#!/usr/bin/env python3

# -*- coding: utf-8 -*-

"""

Created on Thu Jul 10 02:36:26 2025

Bayesian Hierarchical Model using SHOES Pantheon+ Residuals and Planck CMB
TT/TE/EE

Author: Clive Stewart Boyd

Assumeptions...

- SHOES+Pantheon residuals are extracted from observed vs Chebyshev segmented fit

- Planck TT/TE/EE spectra loaded from official text files in Downloads

- Hierarchical priors link AM (angular momentum) evolution to metric perturbations

"""

import numpy as np

import matplotlib.pyplot as plt

from   astropy.io import fits

from   sklearn.preprocessing import StandardScaler

from   scipy.interpolate import interp1d

import pymc as pm

import arviz as az

# === Load SH0ES+Pantheon+ Data ===

y_path = "/Users/boyde/Downloads/ally_shoes_ceph_topantheonwt6.0_112221.fits"

L_path = "/Users/boyde/Downloads/alll_shoes_ceph_topantheonwt6.0_112221.fits"

C_path = "/Users/boyde/Downloads/allc_shoes_ceph_topantheonwt6.0_112221.fits"

with fits.open(y_path) as hdul_y, fits.open(L_path) as hdul_L, fits.open(C_path) as hdul_C:

  y = hdul_y[0].data.astype(np.float64)

  L = hdul_L[0].data.astype(np.float64)
```

```
C = hdul_C[0].data.astype(np.float64)
W = np.linalg.inv(C)
x_best = np.linalg.inv(L @ W @ L.T) @ (L @ W @ y)
mu_fit = L.T @ x_best
mu_obs = y
mu_err = np.sqrt(np.diag(C))
z = np.linspace(0.01, 2.3, len(mu_obs))
z_norm = 2 * (z - z.min()) / (z.max() - z.min()) - 1
# === Load Planck TT, TE, EE Spectra ===
tt_data = np.loadtxt("/Users/boyde/Downloads/COM_PowerSpect_CMB-TT-full_R3.01.txt")
te_data = np.loadtxt("/Users/boyde/Downloads/COM_PowerSpect_CMB-TE-full_R3.01.txt")
ee_data = np.loadtxt("/Users/boyde/Downloads/COM_PowerSpect_CMB-EE-full_R3.01.txt")
Dl_tt = tt_data[:, 1]
Dl_te = te_data[:, 1]
Dl_ee = ee_data[:, 1]
# === Normalize and Resample ===
scaler = StandardScaler()
Dl_tt_norm = scaler.fit_transform(Dl_tt.reshape(-1, 1)).flatten()
Dl_te_norm = scaler.fit_transform(Dl_te.reshape(-1, 1)).flatten()
Dl_ee_norm = scaler.fit_transform(Dl_ee.reshape(-1, 1)).flatten()
N_z = len(z_norm)
x_target = np.linspace(0, 1, N_z)
interp_tt = interp1d(np.linspace(0, 1, len(Dl_tt_norm)), Dl_tt_norm)
interp_te = interp1d(np.linspace(0, 1, len(Dl_te_norm)), Dl_te_norm)
interp_ee = interp1d(np.linspace(0, 1, len(Dl_ee_norm)), Dl_ee_norm)
Dl_tt_resampled = interp_tt(x_target)
```

```
Dl_te_resampled = interp_te(x_target)

Dl_ee_resampled = interp_ee(x_target)


cmb_driver      = (Dl_tt_resampled + Dl_te_resampled + Dl_ee_resampled) / 3.0

# === Residuals ===

residuals = mu_obs - mu_fit

# === Hierarchical Bayesian Model ===

with pm.Model() as hierarchical_model:

    mu_alpha      = pm.Normal("mu_alpha", mu=0, sigma=1)

    mu_beta       = pm.Normal("mu_beta", mu=0, sigma=1)

    alpha         = pm.HalfNormal("alpha", sigma=0.1)

    beta          = pm.HalfNormal("beta", sigma=0.1)

    mu            = mu_alpha + alpha * cmb_driver + beta * z_norm

    sigma_obs     = pm.HalfNormal("sigma_obs", sigma=1)

    obs           = pm.Normal("obs", mu=mu, sigma=sigma_obs, observed=residuals)

    trace         = pm.sample(1000, tune=1000, target_accept=0.9, cores=4,
random_seed=42)

# === Posterior Plot ===

az.plot_trace(trace, var_names=["mu_alpha", "mu_beta", "alpha", "beta", "sigma_obs"])

plt.tight_layout()

plt.savefig("/Users/boyde/Downloads/posterior_hierarchical_am_model.png")

plt.show()
```

Appendix B

Appendix B

```
#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""
Created on Thu Jul 10 06:56:31 2025
#!/usr/bin/env python3
# -*- coding: utf-8 -*-
Bayesian Comparison: Rotating vs Flat STF
Author: Clive Stewart Boyd
"""
import numpy as np
import pymc as pm
import arviz as az
import matplotlib.pyplot as plt
from astropy.io import fits
from sklearn.preprocessing import MinMaxScaler
# === Load SHOES Pantheon+ Data ===
y_path = "/Users/boyde/Downloads/ally_shoes_ceph_topantheonwt6.0_112221.fits"
L_path = "/Users/boyde/Downloads/alll_shoes_ceph_topantheonwt6.0_112221.fits"
C_path = "/Users/boyde/Downloads/allc_shoes_ceph_topantheonwt6.0_112221.fits"
with fits.open(y_path) as hdul_y, fits.open(L_path) as hdul_L, fits.open(C_path) as hdul_C:
    y = hdul_y[0].data.astype(np.float64)
    L = hdul_L[0].data.astype(np.float64)
    C = hdul_C[0].data.astype(np.float64)
mu_obs = y
mu_err = np.sqrt(np.diag(C))
z = np.linspace(0.01, 2.3, len(mu_obs))
```

```
# === Load Planck CMB Spectra ===

cmb_tt = np.loadtxt("/Users/boyde/Downloads/COM_PowerSpect_CMB-TT-full_R3.01.txt")

cmb_te = np.loadtxt("/Users/boyde/Downloads/COM_PowerSpect_CMB-TE-full_R3.01.txt")

cmb_ee = np.loadtxt("/Users/boyde/Downloads/COM_PowerSpect_CMB-EE-full_R3.01.txt")

Dl_tt = cmb_tt[:, 1]

Dl_te = cmb_te[:, 1]

Dl_ee = cmb_ee[:, 1]

l_tt  = cmb_tt[:, 0]

# Normalize and align to z-axis

minlen = min(len(Dl_tt), len(Dl_te), len(Dl_ee), len(z))

scaler = MinMaxScaler()

cmb_driver = (

    scaler.fit_transform(Dl_tt[:minlen, None]).flatten()

    + scaler.fit_transform(Dl_te[:minlen, None]).flatten()

    + scaler.fit_transform(Dl_ee[:minlen, None]).flatten()

) / 3.0

cmb_driver = cmb_driver[:minlen]

mu_obs     = mu_obs[:minlen]

mu_err     = mu_err[:minlen]

z          = z[:minlen]

# === Define Rotating vs Flat STF Hypotheses ===

def rotating_stf(z, alpha, beta):

    return alpha * np.sin(np.pi * z) + beta * z**2

def flat_stf(z, alpha, beta):

    return alpha * z + beta

# === Bayesian Model with Model Comparison ===
```

```
with pm.Model() as model:

    # Priors

    mu_alpha = pm.Normal("mu_alpha", mu=0.01, sigma=0.05)

    mu_beta    = pm.Normal("mu_beta", mu=0.0, sigma=1.0)

    alpha     = pm.HalfNormal("alpha", sigma=0.05)

    beta      = pm.HalfNormal("beta", sigma=1.0)

    sigma_obs = pm.HalfNormal("sigma_obs", sigma=0.5)

    # STF model switch

    is_rotating = pm.Bernoulli("is_rotating", p=0.5)

    # Conditional model based on STF structure

    mu_theory = pm.Deterministic(

        "mu_theory",

        pm.math.switch(

            is_rotating,

            rotating_stf(z, alpha, beta),

            flat_stf(z, alpha, beta),

        ),

    )

    # Likelihood

    pm.Normal("obs", mu=mu_theory + cmb_driver * mu_beta, sigma=mu_err +
sigma_obs, observed=mu_obs)

    trace = pm.sample(2000, tune=1000, target_accept=0.95, cores=4,
return_inferencedata=True)

# === Posterior Summary ===

az.plot_trace(trace, figsize=(12, 10))

plt.tight_layout()

plt.show()

# Model evidence via LOO/WAIC

loo    = az.loo(trace)
```

```
waic = az.waic(trace)

print("\nLOO:", loo)

print("WAIC:", waic)

# Probability of rotating STF

posterior_rot = trace.posterior["is_rotating"].mean().values

print(f"\nPosterior Probability of Rotating STF: {posterior_rot:.3f}")
```